

# Nitidine Reprograms NTERA-2 Cancer Stem Cells: Transcriptomic Insights into EMT, Apoptosis, and Cell-Cycle Activation

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Overview

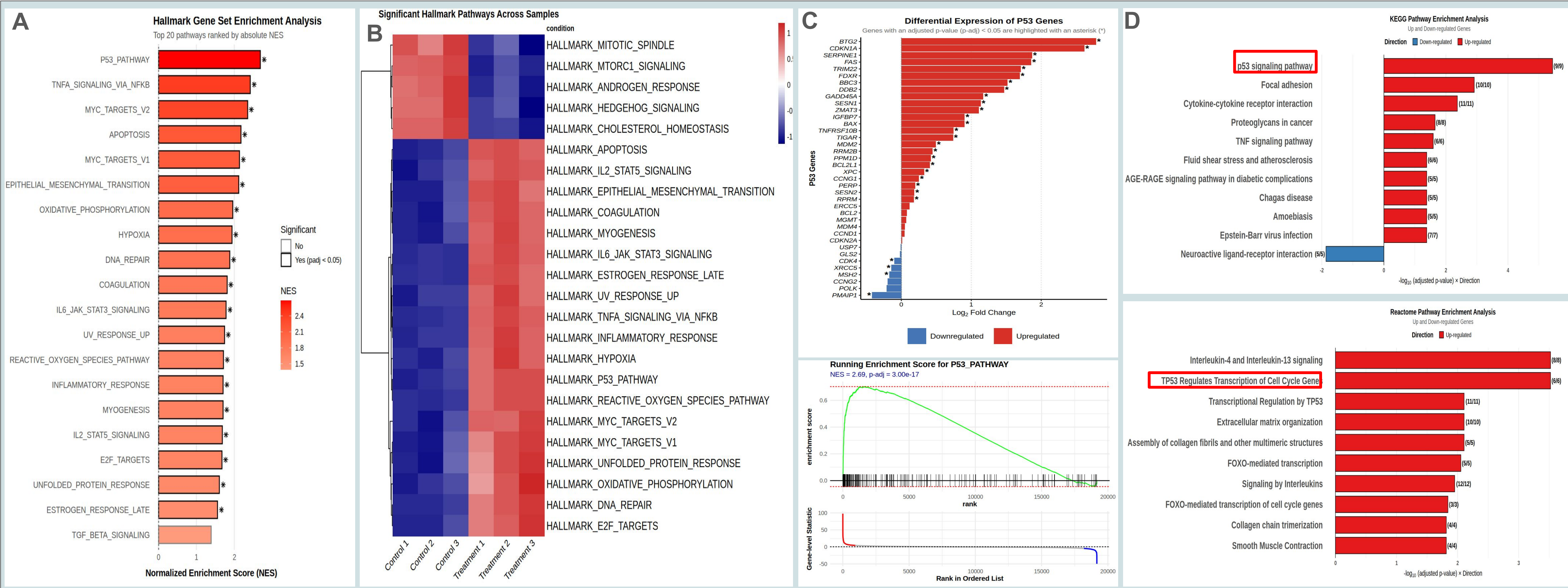
**Background:** Cancer stem cells are recognized as key contributors to tumor malignancy and disease progression. Nitidine has been reported to induce intrinsic apoptosis through activation of the p53 signaling pathway in NTERA-2 cancer stem cells. However, the critical signaling mechanisms underlying the regulation of stemness properties by Nitidine in this cell line have not yet been clearly elucidated.

**Results:** Nitidine reprogrammed the NTERA-2 transcriptome, altering 4,633 genes overall (396 up-regulated; 255 down-regulated). Gene set enrichment indicated robust engagement of p53-dependent apoptosis, with concurrent modulation of Notch and EMT programs.

Method

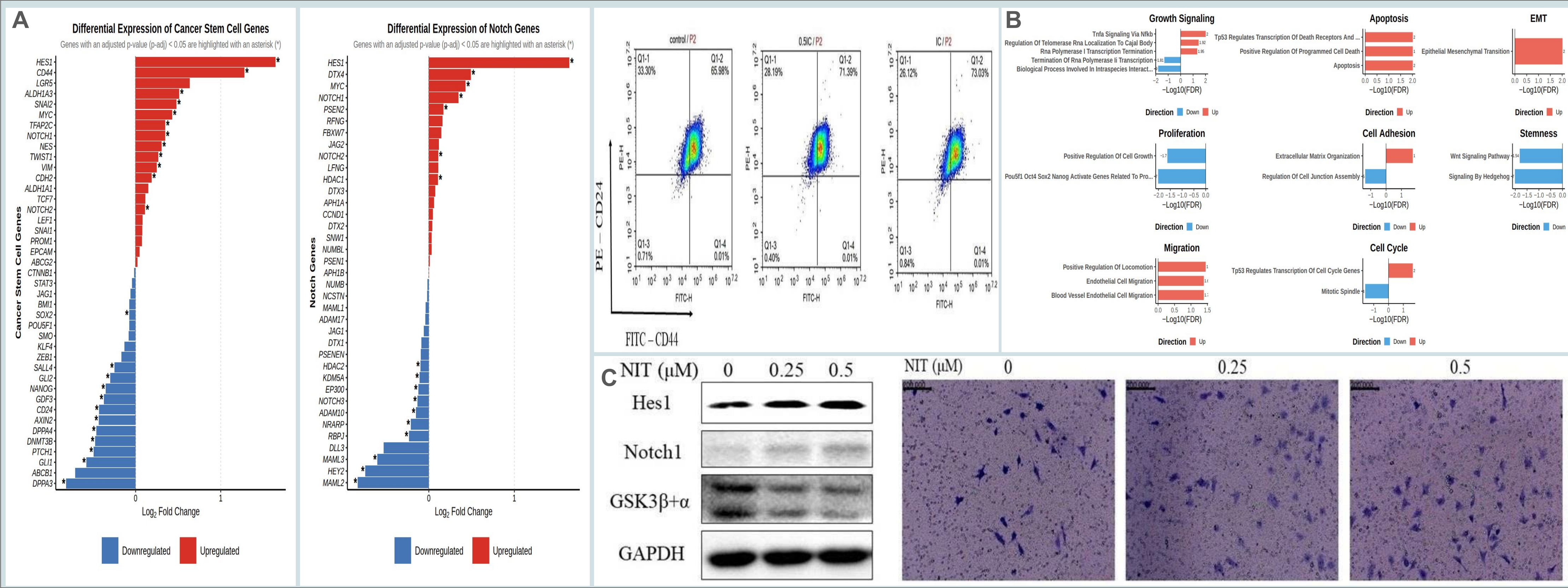
The workflow starts with NTERA-2 cells, divided into 3 Control and 3 Treatment groups. Total RNA is extracted and targets are enriched. The RNA is then fragmented, reverse transcribed, ligated with adapters, and amplified to create a cDNA library. This library is sequenced on a sequencing machine. The resulting data is used for transcriptome/genome mapping, sequencing, and paired-end reads analysis. This leads to tissue-level biological analysis (Volcano plot) and a personalized treatment decision.

Results



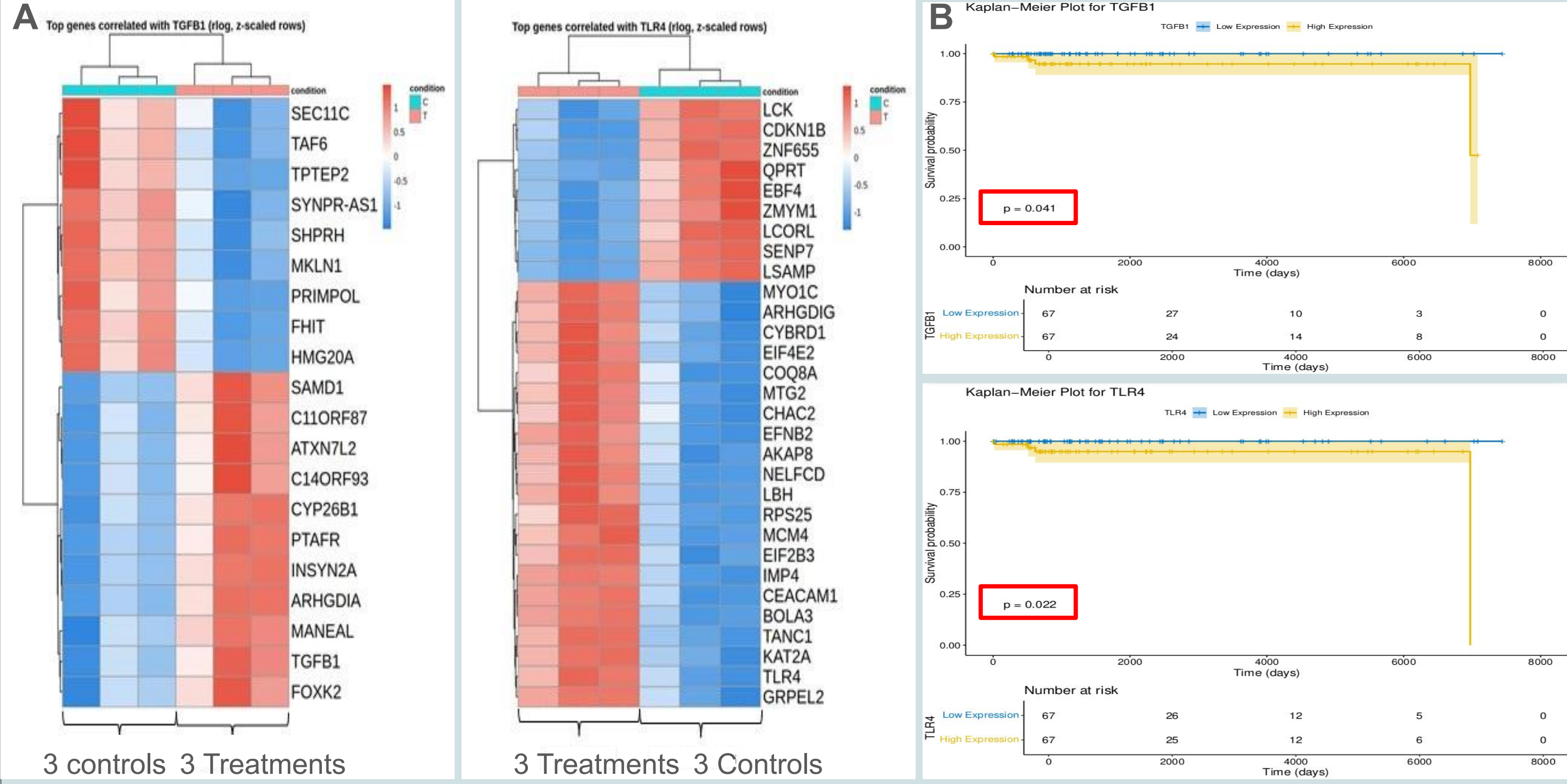
Result 1:  
Nitidine Induces Transcriptomic Reprogramming and Activates Tumor-Suppressive Pathways

**Figure 1. Transcriptomic alterations in nitidine-treated testicular cancer samples.** Hallmark pathway enrichment analysis (A) revealed significant activation of the p53 signaling pathway, TNFα/NF-κB signaling, along with stress-associated programs such as oxidative phosphorylation, and hypoxia. The clustered heatmap (B) showed a clear separation between control (C) and nitidine-treated (T) samples, indicating extensive transcriptional reprogramming. (C) Differential expression analysis of p53 target genes revealed upregulation of canonical mediators including CDKN1A (p21), BTG2, FAS, GADD45A, highlighting activation of DNA-damage response, apoptosis, and cell-cycle arrest programs. Gene set enrichment analysis confirmed robust enrichment of the p53 pathway in treated samples (NES = 2.69, adjusted p =  $3.0 \times 10^{-17}$ ) (D) KEGG and REACTOME pathway enrichment analysis further validated the central role of p53 signaling, which emerged as the top enriched pathway among upregulated genes, alongside related stress



Result 2:  
Nitidine Enhances Pro-Apoptotic Signaling Through Modulation of Key Regulators

**Figure 2. Nitidine reprograms NTERA-2 cancer stem cells by suppressing stemness and Notch signaling while promoting apoptosis and cell-cycle activation.** (A) Transcriptomic profiling revealed broad suppression of cancer stem cell (CSC) genes, including CD44, and pluripotency regulators (POU5F1/OCT4, SOX2, NANOG). Flow cytometry validated this signature, showing an enhancement of the CD44<sup>+</sup>/CD24<sup>-</sup> stem-like fraction after nitidine treatment. (B) Targeted pathway enrichment demonstrated consistent up-regulation of EMT and down-regulation of stemness pathways (Wnt, Hedgehog, Notch), coupled with significant enrichment of apoptosis and cell-cycle-related programs, indicating a shift away from self-renewal toward apoptosis and invasion. (C) Within the Notch pathway, RNA-seq revealed up-regulation of Notch1 and Hes1, which was confirmed by western blot. Functional assays further supported these findings, with nitidine-treated cells showing transformation of cells into more invasive activities, despite reducing stemness markers as well as increasing apoptotic-dependent cell death protein levels and impaired migratory capacity.



Result 3:  
Prognostic of TGFB1 and TLR4 Expression Suggests Therapeutic Synergy with Nitidine

**Figure 3. Prognostic impact of TGFB1 and TLR4 expression.** (A) TGFB1-correlated genes show induction of EMT and proliferation drivers (FOXC2, SAMD1) with loss of tumor suppressors (FHIT, HMG20A), while TLR4 correlates with immune/inflammatory and survival regulators (LCK, CDKN1B, EFNB2). These patterns indicate that nitidine suppresses stemness but promotes TGFB1- and TLR4-driven EMT and inflammatory escape, supporting combination strategies targeting these pathways. (B) In the TCGA testicular cancer cohort, high TGFB1 (p = 0.041) and TLR4 (p = 0.022) expression were both associated with poorer survival, consistent with their roles in immunosuppression and inflammation-driven tumor progression.

Conclusion

Our findings demonstrate that nitidine treatment induces strong transcriptional reprogramming in testicular cancer, activating p53 and apoptosis pathways while suppressing pro-survival signaling. Hallmark enrichment revealed significant upregulation of tumor-suppressive regulators highlighting nitidine's capacity to enhance apoptotic pressure. Survival analyses further strengthen this hypothesis: patients with high TGFB1 or TLR4 expression exhibited significantly worse prognosis, consistent with their roles in immunosuppression and inflammation-driven progression. These results suggest that nitidine's pro-apoptotic activity may be potentiated through combinatorial strategies targeting the TGFB1/TLR4 axis, providing a rational therapeutic framework for overcoming immune evasion in testicular cancer.